

Suitably Impressive Thesis Title



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Abstract

Your abstract goes here. Check your departmental regulations, but this is usually expected to be under 300 words — roughly one page.

A workable structure is one paragraph per move: (1) context and why the problem matters, (2) the gap, (3) what you did, (4) what you found, and (5) what it means more broadly.

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Thank your supervisors first, then anyone whose ideas or feedback shaped the direction of the work.

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Chapter 1

Introduction

One goal of generative modelling is to learn how to produce novel samples that resemble those drawn from an unknown data distribution p_{data} . Diffusion and flow models have been highly successful at this task, achieving state-of-the-art results in image, video and audio generation. Both use the same high-level framework for generation: samples $\mathbf{z} \sim p_{\text{base}}$ drawn from a simple noise distribution are transformed continuously over time towards the data distribution.

Flow and diffusion models specify the dynamics of this continuous transformation between noise and data as governed by an ordinary differential equation (ODE) and a stochastic differential equation (SDE), respectively. An ODE defines a path from a noise to data sample by specifying the local velocity at each point along it. In a flow model, a neural network is trained to predict this velocity. Generating data therefore consists of iteratively advancing a sample from noise to data through a sequence of small steps, querying the neural network at each step to determine the direction of travel. Consequently, although flow models enable high-fidelity generation, they can often require hundreds of neural network evaluations, which can be computationally prohibitive.

Flow maps address this limitation by training a neural network to predict the cumulative displacement between any two times along the noise-to-data paths defined by the ODE, rather than the local velocity at a single point. A flow map can therefore transport a noise sample directly to a data sample in a single neural network evaluation. Flow maps can be learned by bootstrapping from a pre-trained flow model via distillation. However, it is also possible to train a flow map from scratch, as demonstrated by MeanFlow.

1.1 Problem Setting and Research Gap

Many applications require conditional generation: producing samples from the data distribution that satisfy an external condition, such as a class label or text prompt. A canonical example of conditional generation is inverse problem solving, in which an observation $\mathbf{y}$ is generated by corrupting an unknown data sample $\mathbf{x} \sim p_{\text{data}}$ according to $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \boldsymbol{\eta}$, where $\mathcal{A}$ is a known forward operator and $\boldsymbol{\eta}$ represents measurement noise. In image inverse problems, for example, $\mathcal{A}$ may downsample, blur or mask an image. Since these operations typically discard information, multiple data samples may be consistent with the same observation. The goal of inverse problem solving is therefore to draw samples from the posterior $p_{\text{data}}(\mathbf{x} \mid \mathbf{y})$, which represents the distribution of plausible data samples consistent with the given observation $\mathbf{y}$.

One approach to conditional generation is to train a generative model specifically for the conditional task, either from scratch or by fine-tuning a pretrained unconditional model. This can provide fast conditional generation, including one-step generation with a flow map. However, adapting the generative model to each new conditional generation setting typically requires a new round of training or fine-tuning, which can be prohibitively expensive for large generative models.

Guidance-based methods offer an alternative for flow (and diffusion) models. These methods keep the pretrained unconditional flow model frozen but nudge its velocity prediction at each sampling step, thereby steering the path from noise towards a data sample that satisfies the given condition. This approach hence allows for same unconditional model to be reused across different conditional generation settings without fine-tuning.

However, this guidance strategy does not transfer naturally to one-step generation with a flow map. A flow map transports an initial noise sample directly to a data sample in a single evaluation, leaving no intermediate predictions through which guidance can be applied. The key insight of Variational Flow Maps (VFM) is that generation can instead be controlled through the choice of starting noise. For a one-step unconditional flow map, the starting noise is the only remaining lever for controlling its output. Conditional generation can therefore be reframed as learning a distribution over starting noise samples that the flow map sends to data samples satisfying the given condition.

Let $p_{\theta}(\mathbf{x})$ denote the model distribution (which approximates $p_{\text{data}}(\mathbf{x})$) obtained by drawing $\mathbf{z} \sim p_{\text{base}}$ and passing it through the flow map, where θ denotes

the neural-network parameters of the flow map. For a given condition $\mathbf{y}$, VFM formulates the required noise distribution as the noise-space posterior $p_\theta(\mathbf{z} \mid \mathbf{y})$. Intuitively, this distribution is obtained by pulling the corresponding model posterior $p_\theta(\mathbf{x} \mid \mathbf{y})$ backwards through the flow map into noise space.

Although $p_\theta(\mathbf{z} \mid \mathbf{y})$ cannot be sampled from directly, its density can be evaluated up to a normalising constant, making variational inference (VI) a natural approach for approximating it. VI introduces a tractable distribution $q_\phi(\mathbf{z} \mid \mathbf{y})$, called a variational distribution, with parameters ϕ , to approximate this posterior. Fitting $q_\phi(\mathbf{z} \mid \mathbf{y})$ using the variational objective requires both being able to sample from it and evaluate its density. VFM satisfies these requirements by parameterising $q_\phi(\mathbf{z} \mid \mathbf{y})$ as a diagonal Gaussian whose mean and variance are predicted from $\mathbf{y}$ by a neural network, termed the noise adapter.

Although a Gaussian supports both sampling and exact density evaluation, it may be too simple to approximate the potentially complex noise-space posterior $p_\theta(\mathbf{z} \mid \mathbf{y})$. VFM compensates for this by jointly training the noise adapter and the flow map, allowing the geometry of the noise space to be reshaped such that $p_\theta(\mathbf{z} \mid \mathbf{y})$ can be more easily approximated by a simple Gaussian distribution. VFM therefore enables one-step conditional generation, but only by fine-tuning the flow map. This leaves open whether one-step conditional generation can be achieved while keeping the unconditional flow map fixed.

1.2 Proposed Approach and Main Findings

The central question of this project is whether a more expressive variational distribution $q_\phi(\mathbf{z} \mid \mathbf{y})$ can approximate $p_\theta(\mathbf{z} \mid \mathbf{y})$ while leaving the pretrained flow map frozen. We investigate this question by replacing VFM’s Gaussian noise adapter with a conditional normalising flow. Normalising flows support both sampling and exact density evaluation, satisfying the requirements of variational training while being capable of representing substantially more complex distributions than a Gaussian. When amortising over observations $\mathbf{y}$, the inverse-problem setting also naturally permits the approximating distribution, $q_\phi(\mathbf{z} \mid \mathbf{y})$, to be learned via maximum likelihood estimation. We therefore introduce a training objective that combines the variational loss with a maximum-likelihood loss, and show that the combination is more effective than either loss alone.

The resulting approach enables one-step conditional generation using an existing pretrained unconditional flow map while keeping its parameters fixed. Solving

each new conditional task using this approach requires training only a lightweight noise adapter. We evaluate the method across inverse problems of increasing difficulty, progressing from a two-dimensional checkerboard problem to image brightness, $4\times$ super-resolution and inpainting. On the two-dimensional problem, the Gaussian adapter fails to approximate the noise-space posterior accurately, whereas the normalising-flow adapter captures its multimodal structure. For $4\times$ super-resolution, our method achieves the lowest maximum mean discrepancy (MMD) among all evaluated baselines, including VFM.

However, we find that the approach is not competitive on image inpainting. Our analysis suggests that the main difficulty is learning, in a single step, which initial noise samples will map to images matching an observation that constrains many pixel values. This motivates training the adapter to predict a point partway along the noise-to-data path, rather than the initial noise itself, before mapping that point to data in a single flow-map evaluation. We leave the development of this extension to future work.

1.3 Contributions

This thesis makes the following contributions:

1. We adapt the Variational Flow Maps (VFM) framework for one-step conditional generation so that it is training-free with respect to the flow map. In particular we replace the Gaussian noise adapter with a more expressive normalising flow. To our knowledge, this is the first work to investigate solving conditional generation entirely within the noise space of a pretrained unconditional flow map without fine-tuning the flow map itself.
2. We further extend VFM by introducing a training objective that combines both the variational loss with a maximum-likelihood loss. We show that this combined objective is more effective than training the noise adapter with either maximum likelihood or the variational loss alone.
3. We introduce a principled evaluation framework that isolates error attributable to the trained noise adapter from error attributable to the frozen pretrained flow map.
4. We demonstrate that the more expressive normalising flow adapter can substitute for flow map fine-tuning on two-dimensional checkerboard, image-

brightness and $4\times$ super-resolution inverse problems. On super-resolution, our method achieves the lowest MMD among all evaluated baselines, including VFM. However, we find that it is not competitive for image inpainting, and we provide a detailed analysis of its failure.

1.4 Thesis Overview

Chapter 2

Background

2.1 Generative Modelling

The Goal of Generative Modelling

2.1.1 Reward Alignment

2.1.2 Inverse Problems

Relation to Reward Alignment

2.2 Flow-based generative models

2.2.1 Normalising Flows

Constructing Normalising Flows

Training Normalising Flows

Forward KL divergence

Reverse KL divergence

2.2.2 Flow Matching

ODE Definition

Flow Model Definition

Sampling from a Flow Model

Flow Matching

2.2.3 Flow Maps

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2.3 Variational Inference

2.3.1 Amortised Variational Inference

Chapter 3

Related Work

3.1 The Reward-tilted vector field

3.2 Estimating the Value Function with Flow models

Make the reward function robust to noise.

Approximating the posterior as the mean

Monte Carlo approximation of the posterior

Sequential Monte Carlo

3.2.1 Estimating the Value Function with Flow Maps

Chapter 4

Method

4.1 The noise adapter as a normalising flow

4.2 Training objectives

4.2.1 Reverse KL

4.2.2 Forward KL

4.2.3 Mixed Objective

4.2.4 β -VAE

4.2.5 Noise Augmentation

4.3 Building the noise adapter

4.3.1 Low-dimensional Toy Problems

4.3.2 Image-scale Problems

4.4 Evaluation Framework

Additional metrics for Low-dimensional experiments

Image-scale experiments

Chapter 5

Experiments and Results

5.1 2D Checkerboard Example

5.1.1 The Data Distribution

5.1.2 Defining the Inverse Problem

5.1.3 Base Flow Map Training

5.1.4 Noise Adapter Training

Normalising Flow Adapter

Gaussian Adapter with frozen θ

VFM

5.1.5 Results

5.2 Image Inverse Problems

5.2.1 Common Setup

Base Flow Map

TarFlow Noise Adapter Hyperparameters

5.2.2 Brightness Steering

Adapter Training

Results

5.2.3 Super-Resolution

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Noise Adapter Training details

Objective Ablation

Chapter 6

Conclusion

6.1 Summary

6.2 Future Work

6.3 Critical Evaluation

Appendix A

Supplementary Material

A.1 Change of Variables

A.2 Score functions and score matching

A.3 Additional Experiments

A.3.1 GMM Experiments

Capacity Reward Hack in SR Inverse Problem